

User Manual

JapyScope Remote

Version 24.9.2026

A hand controller for GoTo telescope mounts, with a companion app for your phone or computer.



The JapyScope Remote controller.

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PART I

Getting Started

Set up your controller and complete your first alignment.

01 What Is JapyScope Remote?

JapyScope Remote is a hand controller for GoTo telescope mounts. It replaces your mount's original hand controller and connects straight into its hand-control port. Alongside it you get a companion app you can open from your phone, tablet, or computer.

It works with GoTo Alt-Az mounts, Dobsonians included — it's used daily on a Sky-Watcher Flextube 400P.

With it, you can:

- Slew to objects from built-in catalogs, or search for one by name.
- Save your own catalogs and objects from the companion app.
- Track and manually nudge the mount once you're pointed at something.
- Check on the mount and change settings from your phone or computer.

02 The Controller

The controller is a single hand-held unit with three things you'll use constantly:

- **Screen** — shows menus, lists, and status, at the top of the unit.
- **Keypad** — digits 1–9 and 0, plus **FN2** and a delete key (**BKSP**, top right), used for typing and for jogging the mount.
- **Side wheel** — rotate to scroll, push to select, on the right edge of the unit (shown in green below).



The Home screen, with Menu highlighted.

On the back of the unit: a cable to your mount's hand-control port, and a power connector. A cable for an optional external joystick can also be connected there — see Chapter 15.

NOTE

The joystick is an optional accessory. Without one connected, the controller works exactly as described in this manual, using the wheel and keypad alone.

03 Basic Controls

Everything on the controller comes down to the side wheel and the keypad.

Rotate wheel	Scroll up/down through a menu or list.
Push wheel	Open / select / confirm the highlighted row.
Keys 2 / 8	Jog the mount North / South (Up / Down) — on screens where the mount can move: Home, Tracking, and Alignment.
Keys 4 / 6	Jog the mount West / East (Left / Right) — same screens as above.
Key 7	Open the jog-speed adjustment.
Key 9	Go back one screen. Works everywhere, with one exception inside SmartSearch (Chapter 10).
Key 0 / FN2 / BKSP	Digit entry, a small Home-screen surprise, and delete-last-character while typing.

04 Connecting To Wi-Fi

The first time you turn the controller on, it opens its own temporary Wi-Fi network so you can hand it your home Wi-Fi details.

Wi-Fi Setup

Connect to hotspot:

JapyScope-Setup

then open the Web UI · 9=back

Wi-Fi Setup screen.

1. On your phone or laptop, connect to the **JapyScope-Setup** Wi-Fi network. Its password is shown on the controller screen.
2. Open <http://192.168.4.1:8080/setup> in a browser.
3. Enter your home Wi-Fi's name and password, and submit.
4. Reconnect your phone or laptop to your normal home Wi-Fi — the setup network disappears once the controller joins your network.

The Wi-Fi Setup page, open while connected to JapyScope-Setup.

NOTE

Can't find the setup screen? Dial **5000** under Dev Tools (Chapter 14) to open it again at any time.

05 Time, Location, And Backlight

Three quick things to set up before your first night out, all under **Menu** on the controller.

Time Sync

Menu > Time Sync updates the clock over the internet. An accurate clock matters for alignment and slewing.

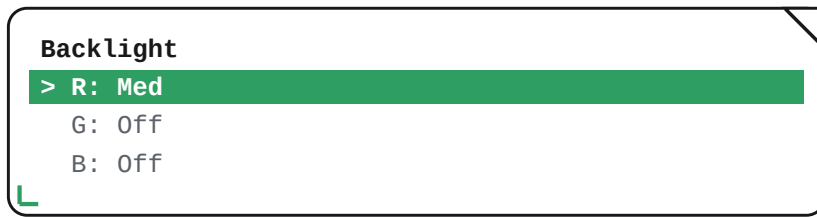
Menu > Time Sync

Location

Menu > Location shows your saved latitude/longitude. Set or update them from the companion app instead — including a one-tap button that fills them in for you (Chapter 19).

Backlight

Menu > Backlight sets the red/green/blue edge lighting. Push the wheel on a colour to cycle it Off, Med, High.



Menu > Backlight

06 Your First Alignment

Go to **Menu > Alignment > Manual** and complete a two-star alignment. This is the same process you'll use any time you realign — it's covered screen by screen in Chapter 9.

NOTE

"Smart" alignment isn't available yet — use **Manual (2 stars)**.

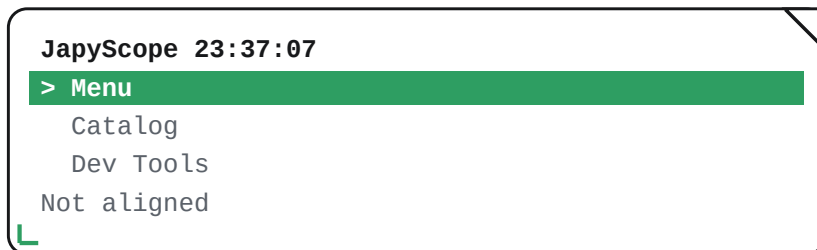
PART II

Using The Hand Controller

Every screen and menu on the controller, and how to use it.

07 The Home Screen

Once you've answered the park question on start-up (below), you land on the Home screen.

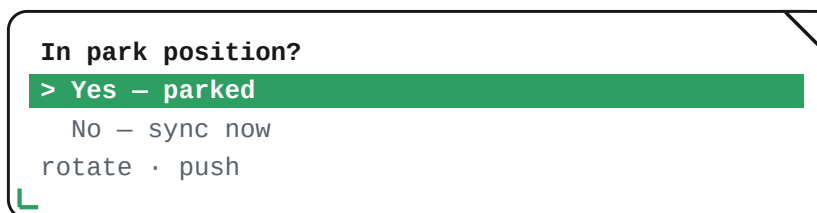


Home screen.

Menu	Settings and utilities — Chapter 8.
Catalog	Find and slew to an object — Chapter 10.
Dev Tools	A code entry for a few extra options — Chapter 14.
Footer	Reads “Not aligned” or “aligned”.
2 / 4 / 6 / 8	Jog the mount directly, no menu needed.
7	Open jog-speed adjustment.

The park question

Every time the controller starts up, it asks whether the mount is in its park position before showing Home.



Start-up park question

- **Yes — parked:** the controller trusts the mount is at its known park position.
- **No — sync now:** the controller notes the mount's current position, so your next slew is still accurate.

If that fails, you'll see a short error screen with the option to try again or go back.

08 The Main Menu

Reached from Home > Menu. Eleven entries, in order:

#	Entry	What it does
1	Time Sync	Update the clock.
2	Alignment	Run a two-star alignment — Chapter 9.

#	Entry	What it does
3	Park / Unpark	Park or unpark the mount.
4	Wi-Fi / Web Access	Get a login code for the companion app.
5	Location	View your saved latitude/longitude.
6	Backlight	Adjust the edge lighting.
7	Language	English or Slovak — applies right away.
8	Interface Theme	How the selected row is marked on screen.
9	Sudo Password	Set a 4-digit password.
10	System	Power off the controller.
11	About	Version information.

9 always leaves the Menu back to Home; the wheel scrolls, push opens the highlighted entry.

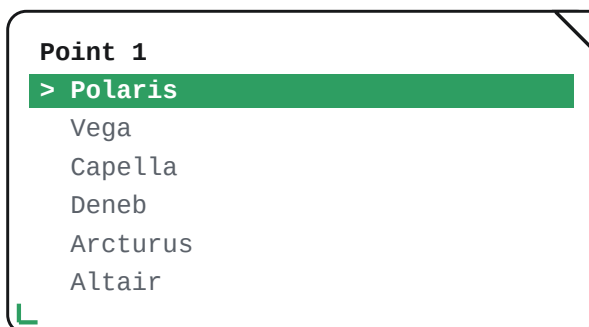
09 Alignment In Detail

Menu > Alignment offers two methods:



Menu > Alignment

Smart isn't available yet. **Manual (2 stars)** is what you'll use:

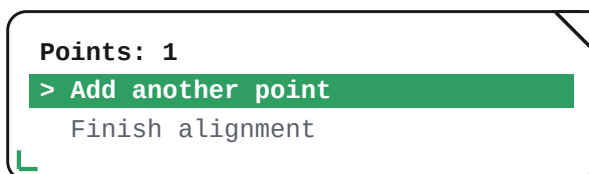


Step 1 — pick a bright star from the list

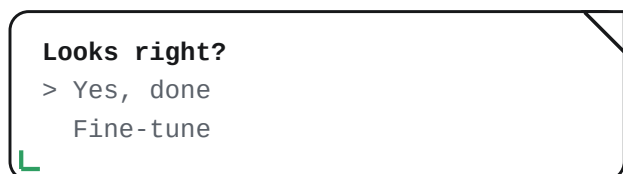


Step 2 — center it using 2/4/6/8 (or the joystick)

Push the wheel once the star is centered. Key 7 opens jog-speed adjustment if the default speed is too fast or too slow.



After each star: add another point, or finish



After the 2nd point: confirm or go back for another

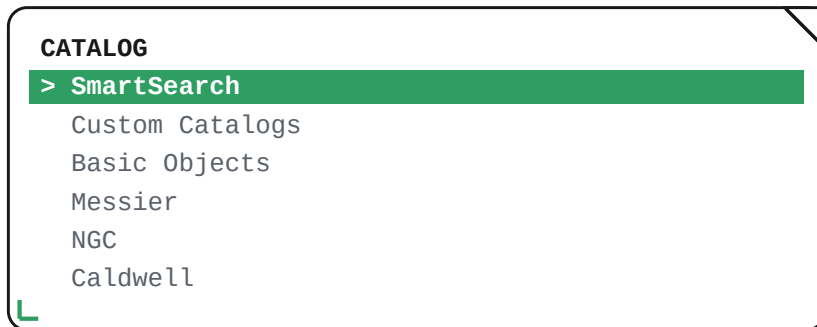
Yes, done saves the alignment — the Home screen's footer updates right away, and future slews won't warn you about it. **Fine-tune** sends you back to pick another star.

NOTE

At any point, 9 backs up one step. From the very first star pick, 9 returns to the Smart/Manual choice.

10 Catalogs And SmartSearch

Home › **Catalog** is how you find something to slew to.

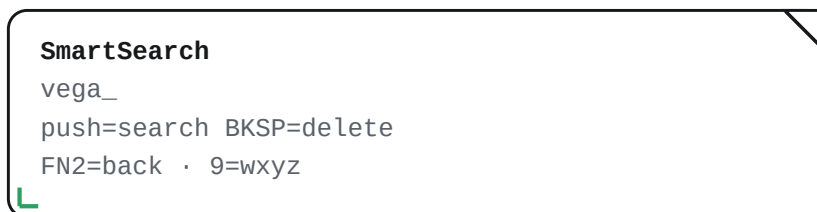


Catalog menu

Entry	Contents
SmartSearch	Type a name and search for it, online and in your catalogs — good for anything not already in a list.
Custom Catalogs	Your own catalogs, created and filled in from the companion app (Chapter 18).
Basic Objects	Moon, Mars, Jupiter, Saturn, Polaris, Vega.
Messier	A small Messier selection.
NGC / Caldwell	A small sample from each.

Typing with SmartSearch

Selecting SmartSearch opens a phone-style typing screen, the same idea as texting on an old phone keypad:



SmartSearch typing

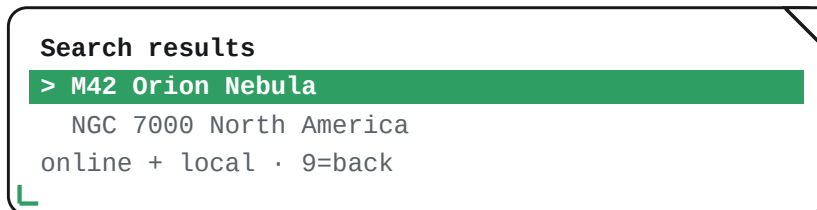
Key	Letters
1	a b c
2	d e f
3	g h i
4	j k l
5	m n o
6	p q r s
7	t u v
8	w x y z
9	w x y z (here only)
0	digit 0 only

NOTE

On this screen only, key **9** types letters instead of going back. Use **FN2** to cancel out of SmartSearch instead. Everywhere else, 9 is always back.

- Tap a key repeatedly, quickly, to cycle its letters.
- **BKSP** deletes the last character; **0** types a 0.
- Push the wheel to search once you've typed something.

Results show both your own catalogs and, when you're online, matches found on the internet:



SmartSearch results

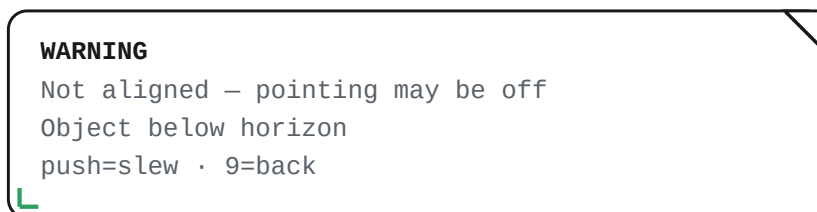
NOTE

No internet? SmartSearch still searches your own catalogs, and says so in the footer.

11 Slewing, Tracking, And Jogging

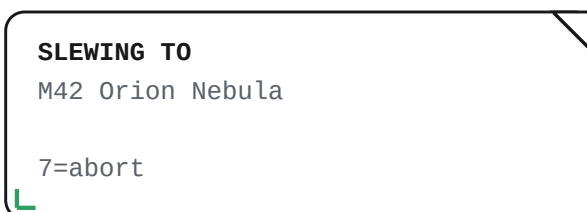
Picking any object — from a catalog or a search result — goes through the same steps.

If you're not aligned, or the object is below the horizon right now, you'll see a warning first:

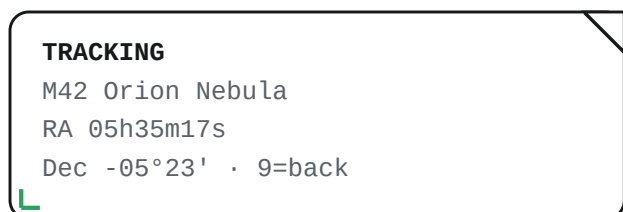


Warning screen

Pushing through the warning (or when there's nothing to flag) starts the slew:

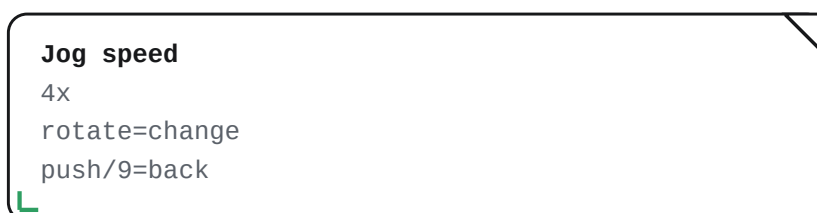


Slewing — key 7 aborts



Tracking, once you arrive

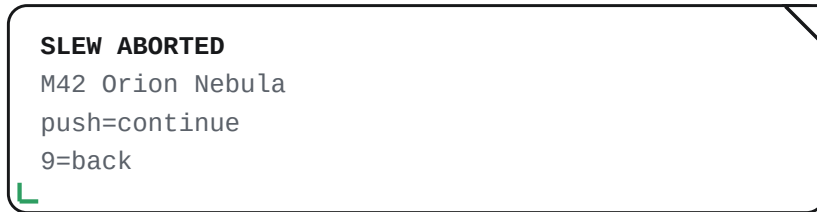
While Tracking, keys 2/4/6/8 (or the joystick) nudge the mount without leaving tracking — handy for fine-centering. Key 7 opens jog-speed adjustment; 9 returns Home.



Jog-speed adjustment

Seven speed steps are available: 1x, 2x, 4x, 8x, 16x, 32x, and 64x.

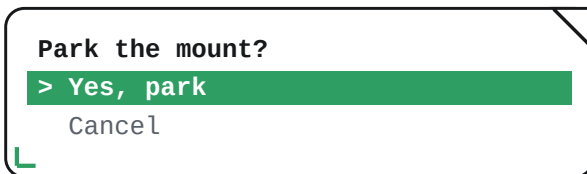
Aborting a slew offers to pick it back up:



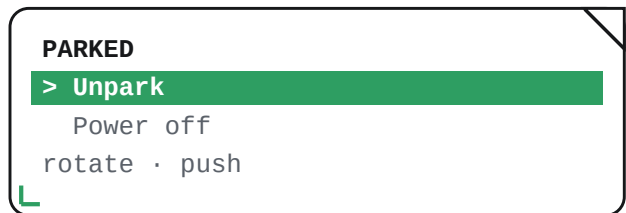
After an aborted slew

12 Parking And Shutdown

Menu > **Park / Unpark** parks or unparks the mount. Parking asks you to confirm first:



Park confirmation



Parked — unpark or power off from here

From **Parked**, **Power off** shuts the controller down cleanly. **Menu** > **System** offers the same power-off, for when you're not already parked.

NOTE

Park the mount before powering off whenever you can — it's how the controller trusts its position again next time it starts.

13 Personalizing The Controller

Language

Menu > **Language** switches between English and Slovak, right away. A few screens are still English only for now.

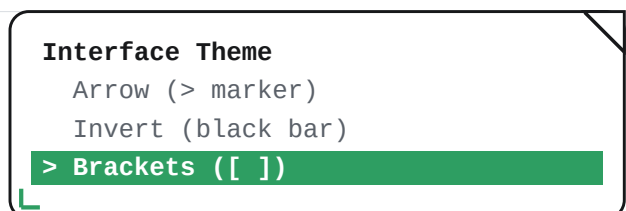
Interface Theme

Menu > **Interface Theme** changes how the selected row of a list is marked:

Theme	How the selected row looks
Arrow (default)	Prefixed with "> ", e.g. "> Menu".
Invert	That row shown in reverse (dark background, light text).
Brackets	Wrapped in [], e.g. "[Menu]".



Picking a theme...



...takes effect right away

Sudo Password

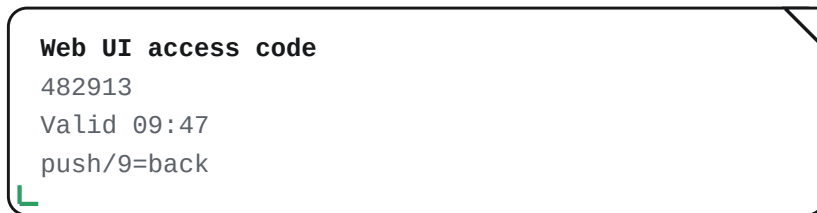
Menu > Sudo Password sets a 4-digit password, dialed in the same way as a Dev Tools code (Chapter 14). The factory default is 1234.

CAUTION

This password is a simple deterrent, not real security. Don't rely on it for anything critical.

Wi-Fi / Web Access

Menu > Wi-Fi / Web Access shows a fresh 6-digit code to log into the companion app, valid for 10 minutes with a countdown:

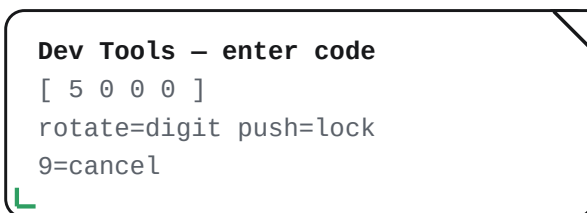


Menu > Wi-Fi / Web Access

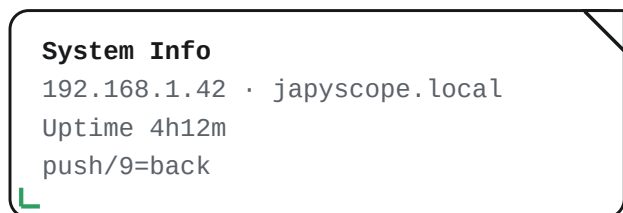
Enter this code on the app's login page (Chapter 16). It expires on its own, and five wrong tries lock you out for 30 seconds.

14 Dev Tools And SSH

Home > Dev Tools opens a 4-digit code entry for a handful of extra options that don't belong in the everyday menu.



Dial each digit, push to lock it in



Example result — code 1234

Rotate the wheel to change the current digit (0–9), push to lock it in and move to the next. After the 4th digit, the code runs. 9 cancels at any point.

Code	What it does
0000	Exit Dev Tools.
0044	Turn on SSH access — see below.
1234	Show basic system info.
5000	Open the Wi-Fi setup screen again.
9955	Run one of your own saved scripts, if you have any.
9999 / 4200 / 1957 / 0905	A few small surprises — find them yourself.
anything else	“Invalid code.”

Terminal commands and SSH

Code **0044** turns on SSH access, so you can connect to the controller from a terminal on your computer. This screen shows the address and username to use.

Once connected, you can generate a Web UI access code from the terminal instead of the controller screen:

```
cd /opt/japyscope/current
sudo -u japyscope env JAPYSCOPE_DEV_MODE=1 .venv/bin/flask \
  --app webui.app:create_app gen-code
```

15 Key Map Quick Reference

Key	Meaning
1, 3	Not used.
2 / 4 / 6 / 8	Jog Up / Left / Right / Down where the mount can move; plain digits (and letters) everywhere else.
7	Open jog-speed adjustment.
9	Go back one screen — except while typing in SmartSearch.
0	Digit “0”.
FN2	A Home-screen surprise; inside SmartSearch, cancels out instead.
BKSP	Delete the last character while typing.
Wheel rotate	Scroll menus and lists.
Wheel push	Open / select / confirm.

Optional joystick

If you've connected one and turned it on in the companion app's Settings, an external joystick is a second way to do the same things:

Joystick input	Behaves as
Push up/down	Wheel rotate (scroll menus/lists).
Push button	Wheel push (select/confirm).
Push left/right	Jog Left/Right, same as keypad 4/6.
Push it straight in one direction — a diagonal push doesn't register, since the mount only moves one direction at a time.	

The Companion App

Check status, manage catalogs, and change settings from your phone or computer.

16 Reaching The App

The companion app runs on the controller itself and opens in any browser on your home network, at port 8080:

- Normally: `http://<device address>:8080/`, from any device on the same Wi-Fi. Find the address under Dev Tools, code **1234** (Chapter 14) — it shows both the controller's IP address and its hostname.
- During Wi-Fi setup only: `http://192.168.4.1:8080/setup`, while connected to JapyScope-Setup (Chapter 4).

NOTE

The controller's hostname depends on how it was named when it was set up, so it won't always be the same on every unit. If a hostname address doesn't load, use the IP address instead — both are shown on the same Dev Tools screen.

Every other page needs a 6-digit code first, shown on the controller (**Menu > Wi-Fi / Web Access**, Chapter 13):

The login page.

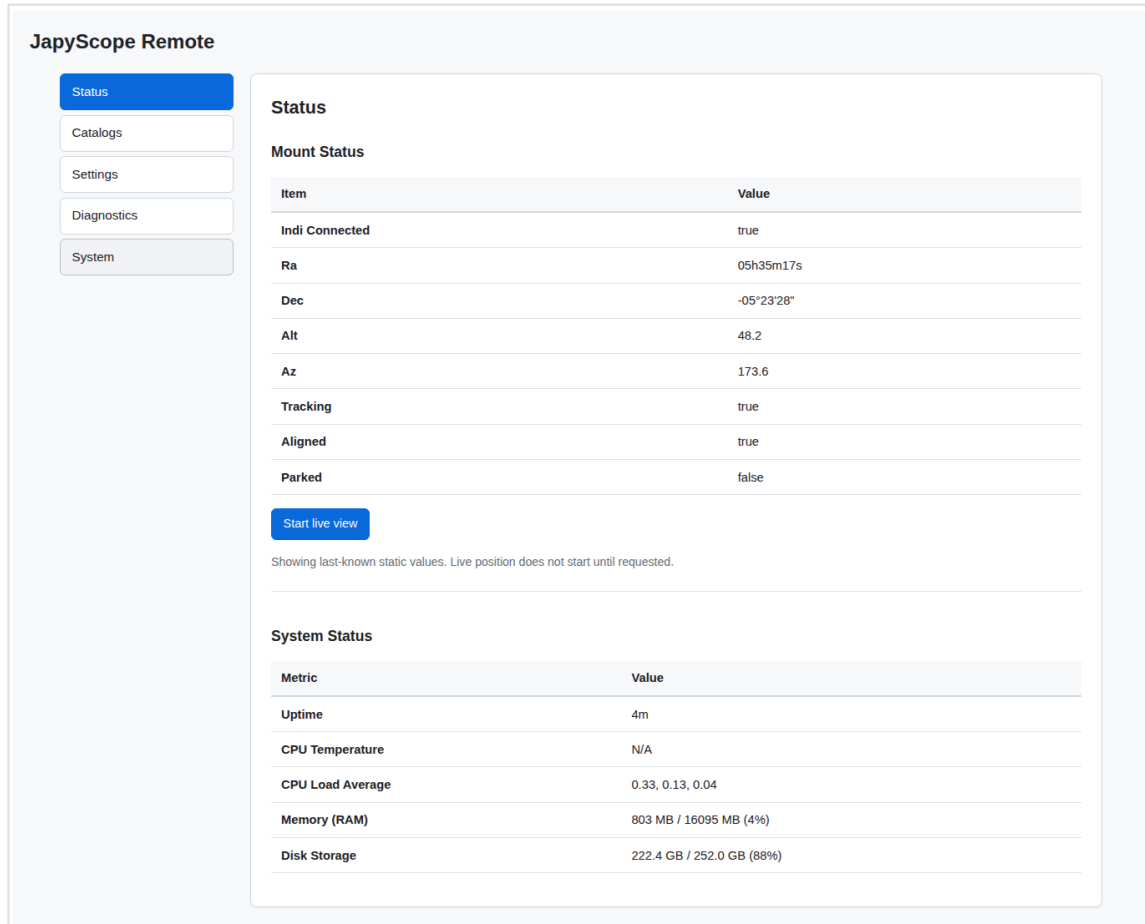
CAUTION

Five wrong codes lock you out for 30 seconds. Codes expire after 10 minutes — get a fresh one on the controller if yours has run out. This is a simple lock, meant for your home network — don't make it reachable from the internet.

Once you're in, a sidebar gives you five pages: **Status**, **Catalogs**, **Settings**, **Diagnostics**, and **System**. None of them can slew, jog, or align the mount — those stay on the controller.

17 Status Page

The page you land on after logging in.



The Status page.

Field	Meaning
Indi Connected	Whether the controller is currently linked to the mount.
Ra / Dec	Where the mount is pointed, in sky coordinates.
Alt / Az	Where the mount is pointed, relative to your horizon.
Tracking	Whether the mount is actively tracking.
Aligned	Whether the current alignment is trusted.
Parked	Whether the mount is at its park position.

NOTE

Position doesn't update on its own. Click **Start live view** to see it refresh every couple of seconds, and **Stop live view** when you're done.

18 Catalogs Page

Create and manage your own catalogs here — these are exactly what shows up under the controller's **Catalog > Custom Catalogs** (Chapter 10). You can have as many as you like.

Status

Catalogs

Settings

Diagnostics

System

Observation Catalogs

New catalog

Create

My Messier Picks

My Messier Picks

Rename

Delete catalog

Name	RA	Dec	Type	Note	Actions
M13	16:41:41	+36:28:00	Cluster	Hercules Cluster	Edit Delete
M31	00:42:44	+41:16:09	Galaxy	Andromeda Galaxy	Edit Delete
M42	05:35:17	-05:23:28	Nebula	Orion Nebula	Edit Delete

Add object

Name

Object name (e.g. M31)

Right Ascension (RA)

HH:MM:SS or deg

Declination (Dec)

±DD:MM:SS or deg

Type

Galaxy, Nebula, Star...

Note

Optional notes

Add object

Import CSV

CSV File

Choose File

No file chosen

Columns: name, ra, dec, type, note · [Download sample CSV](#)

Import CSV

Piconcillo Favorites

Piconcillo Favorites

Rename

Delete catalog

Name	RA	Dec	Type	Note	Actions
NGC 7000	20:58:48	+44:20:00	Nebula	North America Nebula	Edit Delete

Add object

Name

Object name (e.g. M31)

Right Ascension (RA)

HH:MM:SS or deg

Declination (Dec)

±DD:MM:SS or deg

Type

Galaxy, Nebula, Star...

Note

Optional notes

Add object

Import CSV

CSV File

Choose File

No file chosen

Columns: name, ra, dec, type, note · [Download sample CSV](#)

Import CSV

Two catalogs, each with its own objects.

Per catalog

- **Rename / Delete catalog** — deleting a catalog also deletes everything in it.
- **Add object** — give it a name, and optionally its sky coordinates, a type, and a note.
- Each object has **Edit** and **Delete**.

Importing a list

Add many objects at once with **Import CSV**. Columns, in order: name, RA, Dec, type, note — only the name is required. A ready-made sample file is one click away on the page.

NOTE

Have a list in another format (Stellarium, a planetarium export, a plain text list...)? Convert it to this CSV layout first with the catalog import tool at jan-tdy.github.io/useless-astro/tools/catalog-import/.

19 Settings Page

The Settings page.

Location

Latitude, longitude, elevation, and time zone. **Use this device's location** fills these in for you from your browser, when your browser allows it.

Park configuration

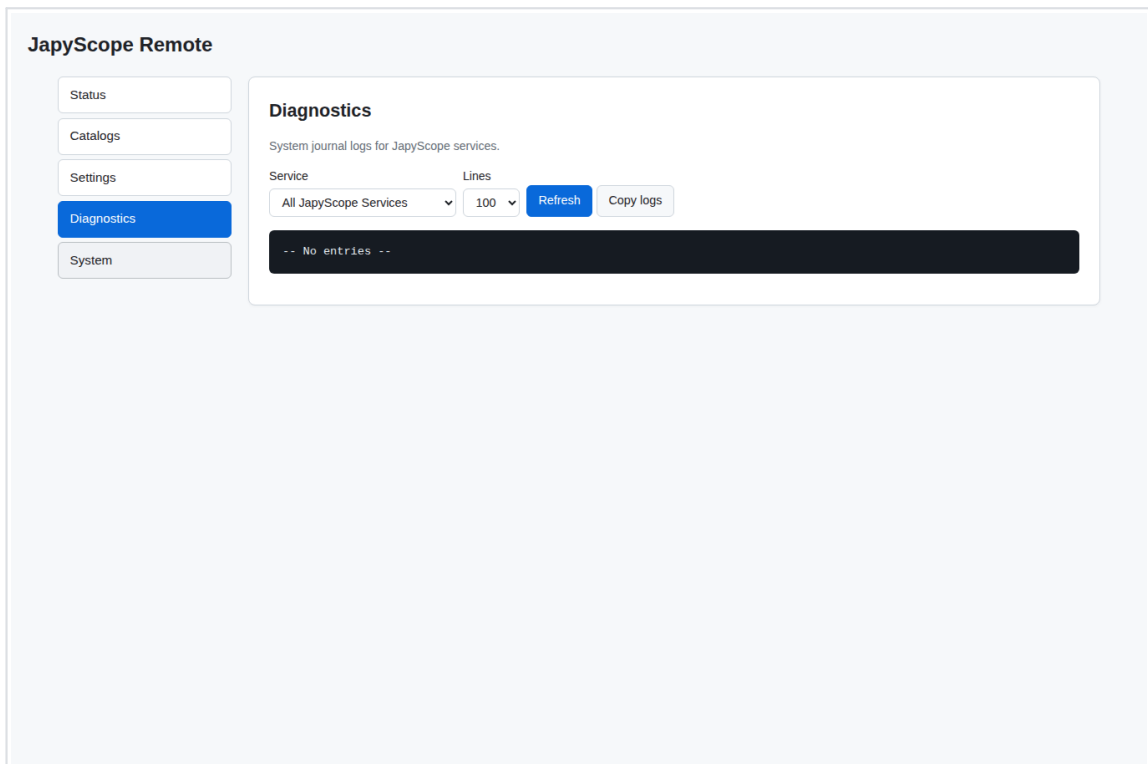
Choose where the mount parks: **zenith**, **home**, or a **custom** position you set yourself.

Joystick

One checkbox, **Joystick connected**, off by default. Turn it on once you've plugged one in — see Chapter 15 for what it does.

This page never touches mount movement or alignment — those stay on the controller.

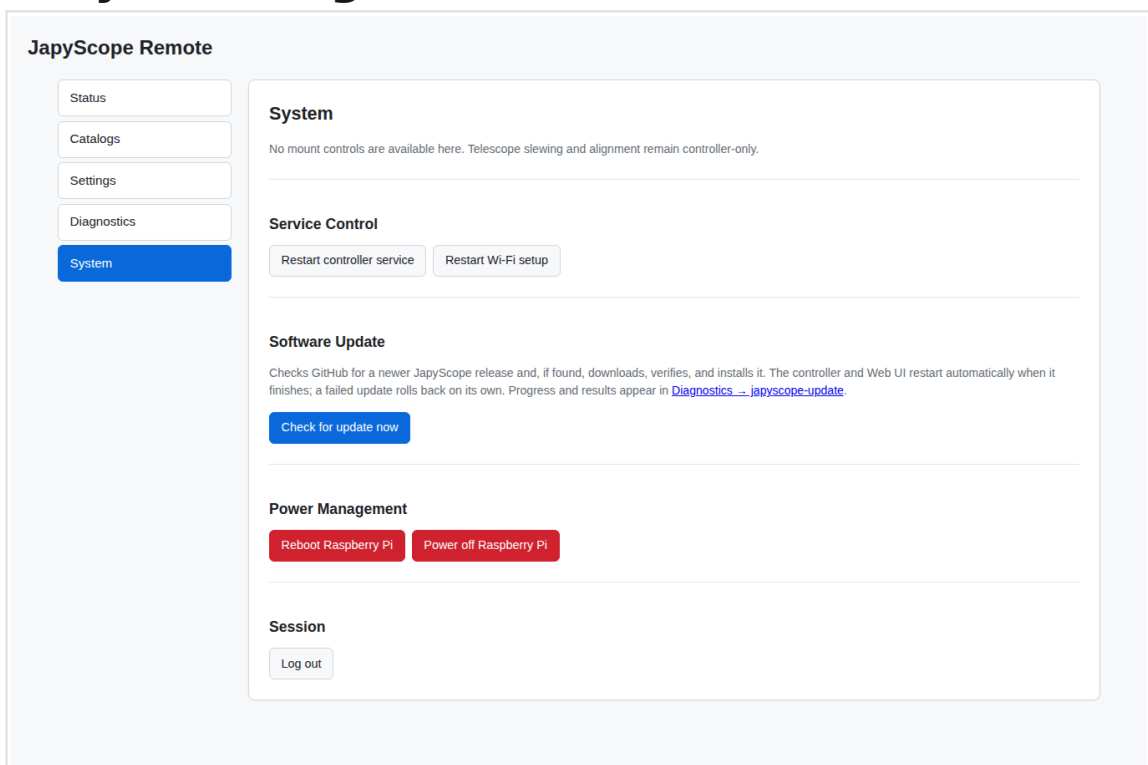
20 Diagnostics Page



The Diagnostics page.

A technical log viewer. You won't normally need it — it's mainly useful if something's wrong and you want to share what happened when asking for help.

21 System Page



The System page.

Button	Effect
Restart controller service	Restarts the controller software, not the whole device.
Restart Wi-Fi setup	Brings the JapyScope-Setup network back up — same as Dev Tools code 5000.
Check for update now	Checks for and installs a software update right away.
Reboot Raspberry Pi	Restarts the whole device.
Power off Raspberry Pi	Shuts the whole device down.
Log out	Ends your session in the app.

22 Security Notes

- The 6-digit login code and the 4-digit sudo password are both simple deterrents, meant for a home network you trust — not real, strong security.
- Nothing here can slew, jog, or align the mount — that stays controller-only, on purpose.

CAUTION

Don't make this app reachable from the public internet. It's built for your home network.

Reference

A few terms, and what to do when something's not working.

23 Glossary

Term	Meaning
GoTo	A mount that can automatically slew to a commanded position.
Alt-Az	A mount that moves up/down and left/right, rather than aligned to Earth's axis.
RA / Dec	An object's fixed position in the sky.
Alt / Az	An object's position relative to your horizon right now — it changes as the sky turns.
Sync	Telling the mount “you are pointed here” without moving it, to fix its known position.
Park	A known, fixed position the mount returns to, so its position is trusted again next time.
SmartSearch	The controller's combined catalog and online object search.

24 Troubleshooting And Support

Problem	What to do
JapyScope-Setup network doesn't appear	Restart the controller. From the app, System > Restart Wi-Fi setup also brings it back.
Locked out of the app (forgot/expired the code)	Get a fresh one on the controller: Menu > Wi-Fi / Web Access.
Mount not responding to slew or jog	Check the cable between the controller and the mount, and that both are powered on.
Controller unresponsive right after power-up	Give it a minute to start up before assuming something's wrong.
An update seems stuck	Check System > Check for update now, and Diagnostics for details.
Still stuck? Get in touch with JapySoft.	